### 2.6.2 ELCB (Earth Leak Circuit Breakers)

An ELCB is a safety device, used in electrical installations with high earth impedance to prevent even minor shock. It detects even small stray voltages on any electrical equipment enclosures and terminates the supply to them. An ELCB is a specialized type of a latching relay that has a provision for incoming mains power connected through its switching contacts. So, the ELCB disconnects the power when earth leakage is detected.




Figure 2.39 Earth Leak Circuit Breakers

The ELCB detects fault currents from live to earth (ground) within the installation and protect the circuit. If sufficient voltage appears across the ELCB's sense coil, it will switch off the power and remain off until manually reset. A voltage sensing ELCB does not sense fault currents from live to any other earthed body. A typical ELCB is shown in Figure 2.39.

## 2.7 SWITCH

These are designed depends upon the contact points (pole) and number of connections (ways).

*A device which is used to connect supply from one point to another point (load) is called a switch.*

There are few type of switches based on requirements.

1. Single Pole Single Way
2. Single Pole Two Way
3. Single Pole Multi Way
4. Double Pole Single Way
5. Double Pole Two Way
6. Double Pole Multi Way
7. Multi Pole Multi Way

Switches are classified based on their voltage and current ratings. In some places separate switches are used for AC and DC. Switches are available based on the following ratings1 A, 2 A, 5 A, 10 A, 15 A, 32 A, 64 A, 100 A and even above. The different types of switches are shown in Figures 2.40(a-g).

We can use switches depends upon our equipment.

1. Normally open
2. Normally closed

Rotary type and slide type switches are also available.

Yet another type called push button switch is available, which makes contact temporarily. This will disconnect the contact while withdrawing the pressure



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